Kernel Concentration and Bandwidth Scaling in Quantum Kernel Methods

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***Abstract*—**

I. Introduction

Quantum kernel methods map classical data into quantum states and use overlaps between states as a kernel for a classical learner, such as a support vector machine (SVM) [1], [2]. Their appeal is that the feature space is an exponentially large Hilbert space. That same size is also a liability: as the number of qubits n grows, the overlaps between states encoding different inputs can become exponentially small and nearly identical for every pair of inputs. When this happens, the kernel matrix approaches the identity and a model trained on it generalizes poorly [3]. Thanasilp *et al.* called this effect *exponential concentration*. They identified expressivity, entanglement, global measurements, and hardware noise as its sources, and they showed that a polynomial number of measurement shots is then not enough to tell kernel values apart [4].

A simple countermeasure is to scale the input features by a bandwidth $\lambda < 1$ before encoding them. Shaydulin and Wild showed numerically that tuning this bandwidth can restore generalization for quantum kernels with many qubits [5], and Canatar *et al.* analyzed how bandwidth controls generalization [6]. Two practical questions remain. First, the kernel must be estimated from a finite number of shots on real hardware, and reducing λ also moves every kernel value toward 1, so there may be no bandwidth at which the kernel is both informative and measurable. Second, a strongly scaled-down quantum kernel may behave much like a classical kernel, so any benefit has to be measured against an equally well-tuned classical baseline.

Research question. How fast do quantum kernel values concentrate as the qubit count grows, and does a bandwidth hyperparameter restore learnability?

Hypotheses. *H1:* at fixed $\lambda = 1$, the off-diagonal kernel variance decays exponentially with n. *H2:* with λ chosen by cross-validation, the kernel keeps enough spread to be measured with 5,000 shots, and its test accuracy stays well above chance and is not worse than a tuned RBF kernel up to $n = 16$.

Falsifier (as stated in our proposal). At $n = 16$ with 5,000 shots per kernel entry, H2 is rejected if (a) at the cross-validated bandwidth λ^* the off-diagonal variance is smaller than the mean shot-noise variance of a single entry, $\overline{k(1-k)}/5000$, or (b) the test accuracy at λ^* is not significantly above chance, or is significantly below the tuned RBF baseline. H1 is rejected if, at $\lambda = 1$, the slope of $\ln \sigma^2$ against n is not negative with a 95% confidence interval excluding zero. Section III-C describes how these tests are applied to our evaluation design.

Contributions. We (i) measure off-diagonal kernel variance and test accuracy from 2 to 16 qubits over a bandwidth grid in exact simulation, (ii) determine where concentration becomes indistinguishable from finite-shot noise, (iii) repeat the variance measurements under a noise model derived from IBM device calibration data and on IBM quantum hardware at selected qubit counts, and (iv) compare every quantum result with an RBF-kernel SVM tuned with the same search protocol.

II. Background

III. Methods

A. Data

We use Fashion-MNIST [7], restricted to the binary pair T-shirt/top (label 0) vs. Pullover (label 2). For each of five seeds we draw a class-balanced subsample of 150 images per class without replacement from the official training split, giving 300 points per seed. To isolate the effect of n and λ from image structure, we also use a synthetic dataset: two isotropic Gaussian classes in $\mathbb{R}^n$ with means $\pm\frac{s}{2}\mathbf{u}$, where $\mathbf{u}$ is a random unit vector and $s = 2$. Its Bayes-optimal accuracy is $\Phi(1) \approx 0.841$ for every n.

B. Folds and preprocessing

The course's NB11 starter notebook was not available to us, so we implemented equivalent fixed folds, leakage-free preprocessing, baselines, ablations, and noise sweeps ourselves. For each seed, five stratified outer folds are generated once, and each outer training set is split into three stratified inner folds. The fold indices are stored on disk, every model reads the same files, and each result records a hash of its fold file. Preprocessing consists of standardization, PCA to n components (Fashion-MNIST only), and a second standardization that gives every component unit variance. It is fit on the training rows of each inner or outer split and only then applied to the held-out rows. A unit test corrupts the held-out rows and checks that neither the fitted transforms nor the selected hyperparameters change.

C. Quantum kernel

Each input $\mathbf{x} \in \mathbb{R}^n$ is encoded on n qubits as $\mathbf{z} = \lambda\mathbf{x}$ using an IQP-type ZZ feature map [1] with $r = 2$ layers and nearest-neighbour entanglement on a line:

$$U(\mathbf{z}) = \prod_{\ell=1}^{r} D(\mathbf{z})\, H^{\otimes n}, \tag{1}$$

$$D(\mathbf{z}) = \exp\Big(i \sum_{j=1}^{n} z_j Z_j + i \sum_{j=1}^{n-1} z_j z_{j+1} Z_j Z_{j+1}\Big). \tag{2}$$

The kernel is the state fidelity $k_\lambda(\mathbf{x}, \mathbf{x}') = |\langle 0^n | U^\dagger(\lambda\mathbf{x}') U(\lambda\mathbf{x}) | 0^n \rangle|^2$. Each layer has n Hadamard gates, n single-qubit R_Z rotations, and $n-1$ two-qubit R_{ZZ} rotations. The compute–uncompute circuit $U^\dagger(\lambda\mathbf{x}')U(\lambda\mathbf{x})$ therefore contains $4(n-1)$ R_{ZZ} gates, for example 60 at $n = 16$. Device-specific depths and two-qubit gate counts after transpilation are reported with the hardware results.

Exact simulation. Because D is diagonal in the computational basis, we compute the statevectors exactly with a batched Walsh–Hadamard transform followed by an elementwise phase. Gram matrices are then computed from inner products of the statevectors. We validate this implementation against Qiskit statevector simulation of the same circuit; the two agree to within 10^{-10} for $n \in \{1, 2, 3, 5\}$ and $\lambda \in \{0.05, 0.7, 2\}$.

Concentration measure. For a Gram matrix K over m points we report the mean and the unbiased sample variance of the $m(m-1)/2$ off-diagonal entries.

Finite-shot estimation. On an ideal device, each entry is estimated as the frequency of the all-zero outcome in N shots of the compute–uncompute circuit, i.e. $\hat{k} = B/N$ with $B \sim \mathrm{Binomial}(N, k)$, so its variance is $k(1-k)/N$. Diagonal entries equal 1 exactly. For $N \in \{10^2, 10^3, 10^4, 10^5\}$ we sample every entry of every training and test Gram matrix independently and repeat the full nested model selection on the sampled matrices, so that the bandwidth and C are chosen under the same shot budget used for testing. We compare the true spread with the shot noise through the ratio $\rho_N = \sigma_\lambda^2 / (\overline{k(1-k)}/N)$, where the bar averages over off-diagonal pairs. We test for any remaining spread with a binomial dispersion test. Under complete concentration, where every off-diagonal value equals a common c, the M sampled off-diagonal entries are i.i.d. $\mathrm{Binomial}(N, c)/N$, so $D = (M-1)s^2/(\hat{c}(1-\hat{c})/N)$ is approximately χ^2_{M-1}. Here s^2 is their sample variance and $\hat{c}$ their mean. Rejecting at $\alpha = 0.05$ means the measured spread cannot be explained by shot noise. The χ^2 reference distribution is a large-sample approximation. It needs no assumption about how the true kernel values depend on each other, because under the null the only randomness is independent shot noise. In 2000 simulated null matrices per setting, the rejection rate at $\alpha = 0.05$ was 0.048 ($c = 0.6$, $N = 500$, $m = 30$), 0.044 ($c = 0.99$, $N = 10^3$, $m = 60$), and 0.051 ($c = 0.95$, $N = 10^4$, $m = 60$).

Falsifier tests. Criterion (a) is evaluated exactly at $N = 5{,}000$ from the saved exact kernel statistics: for each outer fold, the ratio of the off-diagonal variance at λ^* to $\overline{k(1-k)}/5000$ must exceed 1 in more than half of the 25 folds. Our sweep stores shot budgets 10^3 and 10^4 but not 5,000, so λ^* and the accuracies in (b) are taken at both neighbouring budgets. The chance test is a one-sided binomial test of each seed's 300 out-of-fold predictions against 50%, required at $\alpha = 0.05$ for every seed. The RBF comparison is a one-sided Wilcoxon signed-rank test over the 25 paired outer folds ($\alpha = 0.05$); the proposal named McNemar's test, which needs per-sample predictions that our runs did not store. As secondary analyses we also report the pooled dispersion test above and a comparison of λ^* with $\lambda = 1$ (one-sided Wilcoxon over the 25 folds).

Noise model. Noisy simulations use a device model built from a saved calibration snapshot of an IBM backend: per-qubit T_1, T_2, and readout error, and per-gate error and duration for the native gates $\{\mathrm{CZ}, R_Z, \sqrt{X}, X\}$. Each gate applies thermal relaxation over its duration, followed by a depolarizing channel whose strength accounts for the rest of the reported gate error (for $\mathcal{E}(\rho) = (1-p)\rho + pI/d$, the average gate infidelity is $r = p(d-1)/d$). Measurements carry a symmetric readout error. A scale factor s multiplies the depolarizing and readout error rates and divides T_1 and T_2. We use $s \in \{0, 0.5, 1, 2\}$, where $s = 0$ is an ideal device with shot noise only. On test circuits this model agrees with Qiskit Aer's built-in device model to within 0.02 in kernel value. Circuits are transpiled to the backend's coupling map and native gates at optimization level 3 with a fixed seed. For each $n \in \{2, 4, 6, 8, 10\}$ and $\lambda \in \{2^{-4}, 2^{-2}, 2^{-1}, 1\}$, we estimate the kernel between $m = 16$ Fashion-MNIST points (all 120 pairs plus the 16 diagonal circuits) with 4000 shots per circuit. For a set of measured values we report the *excess variance*: the sample variance minus the estimated shot-noise variance $\overline{\hat{k}(1-\hat{k})}/(N-1)$, which is unbiased for the spread of the underlying noisy-device kernel. Its 95% confidence interval comes from bootstrap resampling of points rather than pairs, because pairs that share a point are dependent. The model has known omissions: relaxation acts only during gates (not on idle qubits), and it includes no crosstalk, leakage, or coherent over-rotation errors, so it is expected to underestimate the noise of a real device.

Hardware. All hardware runs used `ibm_marrakesh` on 2026-11-11 through the Qiskit Runtime Sampler, with circuits transpiled against the device's live target at optimization level 3 (fixed transpiler seed). Each compute–uncompute circuit places a barrier between $U(\lambda\mathbf{x}_i)$ and $U^\dagger(\lambda\mathbf{x}_j)$; without it the transpiler cancels the diagonal circuits ($i = j$) down to a bare measurement, so they would probe readout error only. The main run (job `dbq68tgqtr0c738fd5qg`) covered $n \in \{2, 4, 6\}$, $\lambda \in \{2^{-4}, 2^{-2}, 1\}$, and $m = 10$ points per block (45 pairs and 10 diagonals): 495 circuits at 1,000 shots, charged 135 s of QPU time. Before submission, a local preflight ran the exact saved circuits under an ideal and a calibrated noisy simulator and checked circuit counts, agreement with the exact kernel, diagonal values, and the projected QPU time. A 21-

circuit pilot (job `dbq5udoou4as73cukgg0`, 8 s, $n = 2$, $\lambda = 1$, $m = 6$) measured the cost per circuit beforehand; it predates the barrier, so its diagonal values reflect readout error only. The same ISA circuits were then sampled with the same number of shots under the noise model built from the calibration in effect when the job ran, so device and model are compared on identical circuits, qubits, and shot counts. Table I lists the circuit depths and two-qubit gate counts.

D. Model selection and baselines

All kernel models are SVMs with precomputed kernels. Hyperparameters are chosen by nested cross-validation: for each outer fold, the pair (h, C), with h the kernel bandwidth and $C \in \{0.1, 1, 10, 100\}$, is selected by mean inner-fold validation accuracy. The model is then refit on the full outer training set and scored once on the outer test set. The quantum bandwidth is searched over $\lambda \in \{2^{-10}, 2^{-9}, \ldots, 2^{1}\}$ (12 values, including the unscaled $\lambda = 1$). The RBF baseline $k(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$ uses $\gamma = g/n$ with $g = \lambda^2$ over the same 12 values of λ. For small λ the fidelity kernel behaves as $1-O(\lambda^2\|\mathbf{x}-\mathbf{x}'\|^2)$, so this pairs the two grids point by point. Both searches use the same grid size and the same selection protocol. Because each component has unit variance, $g = 1$ corresponds to scikit-learn's default `gamma="scale"`. A first run with narrower grids selected the lowest value in 17–19% of folds overall (up to 36% at some qubit counts) and the highest value in none, so the grids were extended downward before the main experiments. Further baselines are a linear-kernel SVM, ℓ_2-regularized logistic regression with C tuned on the same grid, and the majority-class rate. We report mean test accuracy with 95% confidence intervals over the five seeds, averaging the five outer folds within each seed.

E. Deviations from the proposal

Our proposal planned a single split of 1,200 training and 200 test samples, an IQP map with all-to-all ZZ terms, a second (Hamiltonian-evolution) feature map, a PSD projection of noisy kernel matrices, and 5,000 shots. The study differs as follows. (i) *Evaluation:* we use 300 class-balanced points from the official training split for each of five seeds, with nested cross-validation (5 outer $\times$ 3 inner folds), so that every point is tested once and results carry confidence intervals across subsamples; this also keeps the cost of the $n = 16$ shot-sampled sweeps manageable. (ii) *Entanglement:* ZZ terms couple nearest neighbours only, which keeps hardware circuits shallow on IBM's heavy-hex connectivity. (iii) *Second feature map:* the Hamiltonian-evolution map was not implemented; ablations instead vary the number of layers, remove entanglement, and change the training-set size. (iv) *Shots:* kernel entries are sampled exactly from the binomial distribution for $N \in \{10^2, \ldots, 10^5\}$, and model selection is repeated under each budget; no PSD projection is applied, since SVM training with a precomputed kernel does not require it. (v) *Grids:* $C \in \{0.1, 1, 10, 100\}$ and $\lambda \in \{2^{-10}, \ldots, 2^{1}\}$ instead of the proposal's C grid and $\lambda \in [10^{-3}, 1]$. (vi) *Falsifier tests:* adapted to this design as described above.

IV. Results

V. Discussion

VI. Conclusion

Author Contributions

References

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TABLE I

RESOURCE COUNTS PER KERNEL CIRCUIT. EXACT SIMULATION: LOGICAL COMPUTE–UNCOMPUTE CIRCUIT (λ-INDEPENDENT STRUCTURE). NOISE MODEL AND HARDWARE: TRANSPILED CIRCUITS AS EXECUTED, MIN/MEDIAN/MAX OVER ALL CIRCUITS FOR THAT n (CZ = NATIVE TWO-QUBIT GATE).

Setting	n	Qubits	Depth	2Q gates	Shots
exact simulation (logical circuit)	2	2	12	4 RZZ	exact; 10^2–10^5 sampled
exact simulation (logical circuit)	4	4	18	12 RZZ	exact; 10^2–10^5 sampled
exact simulation (logical circuit)	8	8	26	28 RZZ	exact; 10^2–10^5 sampled
exact simulation (logical circuit)	12	12	34	44 RZZ	exact; 10^2–10^5 sampled
exact simulation (logical circuit)	16	16	42	60 RZZ	exact; 10^2–10^5 sampled
noise model (ibm_marrakesh)	2	2	11/24/26	0/4/4 CZ	4000
noise model (ibm_marrakesh)	4	4	24/87/92	4/24/24 CZ	4000
noise model (ibm_marrakesh)	6	6	71/117/123	20/40/40 CZ	4000
noise model (ibm_marrakesh)	8	8	84/149/158	40/56/56 CZ	4000
noise model (ibm_marrakesh)	10	10	87/179/187	48/72/72 CZ	4000
hardware hw1 (ibm_marrakesh)	2	2	24/24/29	4/4/4 CZ	1000
hardware hw1 (ibm_marrakesh)	4	4	30/87/96	8/24/24 CZ	1000
hardware hw1 (ibm_marrakesh)	6	6	71/117/126	20/40/40 CZ	1000
hardware pilot (ibm_marrakesh)	2	2	1/20/20	0/3/3 CZ	1000